

Fubini's Thm: if f is cts, then you can switch the order for x & y .

Iterated Integral.

Double integrals

$$\iint_R f(x) dA = \int_a^b \int_c^d f(x) dy dx = \int_c^d \int_a^b f(x) dx dy$$

- Find integrals with 1 variable at a time (considering all other variables as constants)
- Evaluate the definite integral before looking at the next one.
- All rules apply (u-substitution, integration by parts)
- Meaning: Volumes above an area.

Double integral on an irregular area

- Make plots!
- Find bounds for x and y
- Bounds w/ variables cannot occur at the outer integrals!

The net volume

below a curve $f(x,y)$ and above region R :

$$\iint_R f(x,y) dA$$

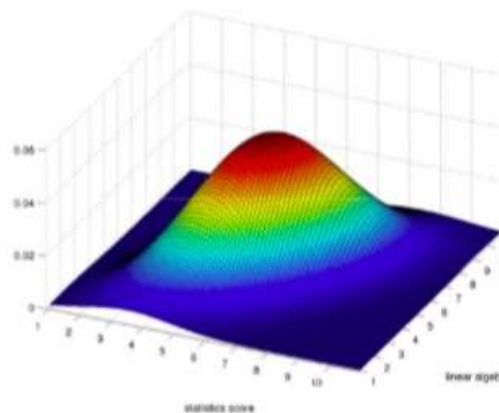
If f is a density function of a variable, then, $\iint_R f(x,y) dA$ gives a quantity of the amount of this variable over the region R

The area of a region R :

$$\iint_R 1 dA$$

Average value of f :

$$\bar{f} = \frac{\iint_R f dA}{\iint_R dA}$$



Polar Integral

Changing Cartesian Integral into Polar Integral

$$\iint_R f(x,y) dx dy = \iint_G f(r \cos(\theta), r \sin(\theta)) r dr d\theta$$

$$x = r \cos(\theta), \quad y = r \sin(\theta)$$

Area in Polar Coordinates

$$A = \iint_R r dr d\theta.$$

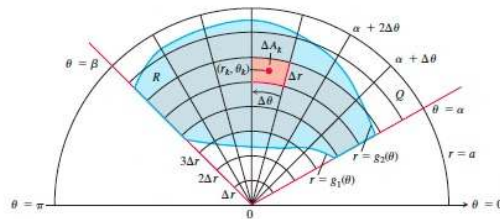
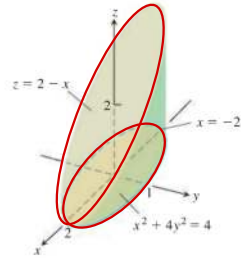


FIGURE 15.22 The region $R: g_1(\theta) \leq r \leq g_2(\theta)$, $\alpha \leq \theta \leq \beta$, is contained in the fan-shaped region $Q: 0 \leq r \leq a$, $\alpha \leq \theta \leq \beta$, where $0 \leq \beta - \alpha \leq 2\pi$. The partition of Q by circular arcs and rays induces a partition of R .

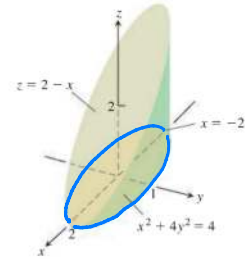
Triple Integral w/ variables in bounds

Inner Integral: two surfaces bounding the volume
e.g. expressions z can take in terms of x & y .

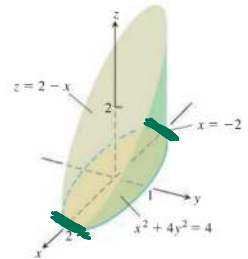
$$z \in [0, 2-x]$$



Middle Integral: two curves bounding the "base"
e.g. expressions y can take in terms of x
 $y \in [-\frac{1}{2}\sqrt{4-x^2}, \frac{1}{2}\sqrt{4-x^2}]$



Outer Integral: two values bounding the "base"
e.g. values x can take
 $x \in [-2, 2]$



Center of mass, Moment & Moment of Inertia.

Given the density function $\delta(x, y, z)$

Mass: $M = \iiint_D \delta(x, y, z) dV$

Center of mass : $(\bar{x}, \bar{y}, \bar{z})$ where

$$\bar{x} = \frac{1}{M} \iiint_D x \delta(x, y, z) dV$$
$$\bar{y} = \frac{1}{M} \iiint_D y \delta(x, y, z) dV$$
$$\bar{z} = \frac{1}{M} \iiint_D z \delta(x, y, z) dV.$$

first Moment about yz -plane : $M_{yz} = M \cdot \bar{x}.$

first Moment about xz -plane : $M_{xz} = M \cdot \bar{y}.$

first Moment about xy -plane : $M_{xy} = M \cdot \bar{z}.$

Moment of Inertia about x axis : $I_x = \iiint_D \underbrace{(y^2 + z^2)}_{\text{distance from a point to } x\text{-axis}} \delta(x, y, z) dV$

Moment of Inertia about y axis : $I_y = \iiint_D (x^2 + z^2) \delta(x, y, z) dV$

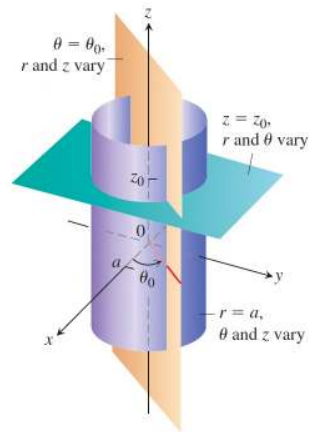
Moment of Inertia about z axis : $I_z = \iiint_D (x^2 + y^2) \delta(x, y, z) dV$

Equations Relating Rectangular (x, y, z) and Cylindrical (r, θ, z) Coordinates

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z,$$
$$r^2 = x^2 + y^2, \quad \tan \theta = y/x$$

$$\iiint_D f \, dV = \iiint_D f \, dz \, r \, dr \, d\theta.$$

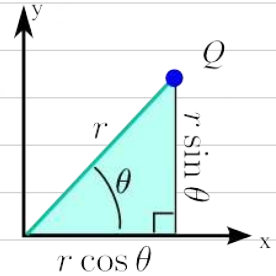
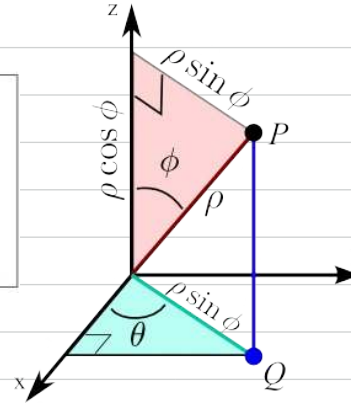
$$\text{Volume} = \iiint_D 1 \, dz \, r \, dr \, d\theta.$$



Equations Relating Spherical Coordinates to Cartesian and Cylindrical Coordinates

$$\begin{aligned} r &= \rho \sin \phi, & x &= r \cos \theta = \rho \sin \phi \cos \theta, \\ z &= \rho \cos \phi, & y &= r \sin \theta = \rho \sin \phi \sin \theta, \\ \rho &= \sqrt{x^2 + y^2 + z^2} = \sqrt{r^2 + z^2}. \end{aligned}$$

(1)



$$\iiint_D f(\rho, \phi, \theta) dV = \iiint_D f(\rho, \phi, \theta) \rho^2 \sin(\phi) d\rho d\phi d\theta$$

$$\text{Volume} = \iiint_D \rho^2 \sin(\phi) d\rho d\phi d\theta.$$

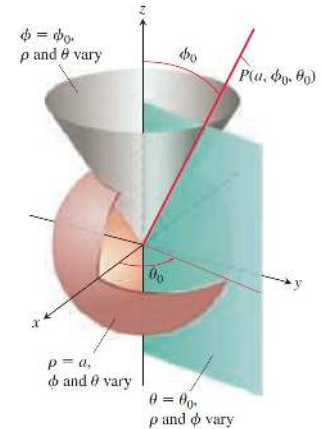


FIGURE 15.52 Constant-coordinate equations in spherical coordinates yield spheres, single cones, and half-planes.